Border rank two does not force a tensor-square rank saving

Proposed resolution; pending independent mathematical review.

Abstract

An explicit smooth, irreducible, nondegenerate curve $X \subset \mathbb{P}^{11}$ and a point p satisfy $\underline{\mathrm{R}}_X(p) = 2$, $\mathrm{R}_X(p) = 3$, and $\mathrm{R}_{X\times X}(p \otimes p) = 9$. Thus the implication in repository problem TR-27 is false for its stated class of projective varieties. More generally, for every $r \geq 3$ and every $m \geq 2$, a smooth projected rational normal curve has a point of border rank two and rank r whose first m tensor powers have ranks $r, r^2, \ldots, r^m$. The proof is elementary: a projection preserves independence of small sets of distinct curve points while identifying a high-rank tangent vector with a sum of r curve points. All tensor products retain their separate factors.

1 The claim being answered

For a reduced irreducible nondegenerate projective variety $X \subset \mathbb{P}(V)$ over $\mathbb{C}$, let $\mathrm{R}_X(p)$ be the smallest number of points of X whose span contains p. Border rank $\underline{\mathrm{R}}_X(p)$ is defined by Zariski closure of these rank loci. Scalars in linear combinations are unrestricted. For $k \geq 1$, write $X^{\times k}$ for its Segre-embedded product in $\mathbb{P}(V^{\otimes k})$.

Problem TR-27 asks whether

$$\underline{\mathrm{R}}_X(p) < \mathrm{R}_X(p) \quad \Longrightarrow \quad \mathrm{R}_{X\times X}(p \otimes p) < \mathrm{R}_X(p)^2 \tag{1}$$

always holds. The same implication is Conjecture 1.1 of Ballico–Bernardi–Gesmundo–Oneto–Ventura [2], and is restated as Conjecture 4.9 in [3]. The construction below disproves precisely this general-variety statement. It does not assert a counterexample when X is required to be a Segre or a Veronese variety.

Theorem 1. *Let $r \geq 3$ and $m \geq 2$ be integers, and put $D = r^m + r$. There are a smooth rational nondegenerate curve $X \subset \mathbb{P}^{D-1}$ and a point p such that*

$$\underline{\mathrm{R}}_X(p) = 2, \qquad \mathrm{R}_X(p) = r, \qquad \mathrm{R}_{X^{\times k}}(p^{\otimes k}) = r^k \quad (1 \leq k \leq m).$$

In particular, $r = 3$, $m = 2$, and $D = 12$ give a counterexample to (1).

2 Two elementary independence lemmas

Let $E_D = \mathbb{C}^{D+1}$ with basis $e_0, \ldots, e_D$. Write

$$c_D(t) = \sum_{j=0}^{D} t^j e_j \quad (t \in \mathbb{C}), \qquad c_D(\infty) = e_D.$$

The projective points $[c_D(t)]$ form the rational normal curve C_D. Distinct points of C_D are linearly independent whenever their number is at most $D + 1$. This follows from the Vandermonde determinant, including the usual last-coordinate version when one point is at infinity.

Lemma 2. *For $D \geq 2$, $\mathrm{R}_{C_D}([e_1]) = D$.*

Proof. Suppose e_1 were a linear combination of $\ell \leq D-1$ distinct curve vectors, with parameter set $S \subset \mathbb{P}^1$. Define

$$f(T) = T \prod_{a \in S \setminus \{0,\infty\}} (T-a).$$

Then $f'(0) \neq 0$. Every finite parameter in S is a root of f. If $\infty \notin S$, then $\deg f \leq \ell + 1 \leq D$; if $\infty \in S$, then $\deg f \leq \ell \leq D-1$. Write $f(T) = \sum_{j=0}^{D} f_j T^j$. The linear functional $L_f(e_j) = f_j$ annihilates every curve vector in the putative decomposition: for finite a, $L_f(c_D(a)) = f(a)$, while $L_f(c_D(\infty)) = f_D = 0$ when infinity occurs. But $L_f(e_1) = f_1 = f'(0) \neq 0$, a contradiction.

For the upper bound, summation over the Dth roots of unity gives

$$e_1 = \frac{1}{D} \sum_{\zeta^D = 1} \zeta^{-1} c_D(\zeta).$$

Indeed, for $0 \leq j \leq D$, the sum of ζ^{j-1} is nonzero only at $j = 1$. □

Lemma 3. *Suppose that every set of at most K distinct points of $X \subset \mathbb{P}(V)$ is linearly independent. If $\mathrm{R}_X(p) = r$ and $r^k - 1 \leq K$, then*

$$\mathrm{R}_{X^{\times k}}(p^{\otimes k}) = r^k.$$

Proof. An r-term expression for p gives the upper bound by expansion. Suppose, for a contradiction, that a representative v of p satisfies

$$v^{\otimes k} = \sum_{\ell=1}^{L} \lambda_\ell x_{1\ell} \otimes \cdots \otimes x_{k\ell}, \qquad L \leq r^k - 1, \tag{2}$$

where $[x_{j\ell}] \in X$. For each factor j, collect the distinct projective points that occur in that factor and choose one representative for each. Denote this set of representatives by A_j. It has at most $L \leq K$ elements, so is linearly independent.

Contract (2) in all factors except j with any functional $\varphi \in V^*$ satisfying $\varphi(v) = 1$. It follows that $v \in \operatorname{span} A_j$. In the unique expansion of v in the basis A_j, at least r coefficients are nonzero, since otherwise $\mathrm{R}_X(p) < r$.

The products of the bases $A_1, \ldots, A_k$ form a basis of their tensor-product span. In this basis, $v^{\otimes k}$ has at least r^k nonzero coordinates: its support is the Cartesian product of the nonzero supports of the k expansions of v. The right-hand side of (2) uses at most L such basis vectors, even after repeated products are combined. This is impossible for $L < r^k$. □

3 The projected curve

Choose pairwise distinct nonzero complex numbers $a_1, \ldots, a_r$; the choices $a_i = i$ suffice. Set

$$b = \sum_{i=1}^{r} c_D(a_i), \qquad z = e_1 - b, \qquad V = E_D / \mathbb{C}z,$$

and let $\pi : E_D \to V$ be the quotient map. Lemma 2 and subadditivity of rank give

$$D = \mathrm{R}_{C_D}([e_1]) \leq \mathrm{R}_{C_D}([z]) + \mathrm{R}_{C_D}([b]) \leq \mathrm{R}_{C_D}([z]) + r. \tag{3}$$

Consequently $\mathrm{R}_{C_D}([z]) \geq D - r = r^m$; in particular, $z \neq 0$ and $[z] \notin C_D$. The restriction of projective projection to C_D is therefore a morphism. Let X be its image, with its reduced structure. It is closed and irreducible because C_D is projective and irreducible. It is nondegenerate because C_D spans $\mathbb{P}(E_D)$ and π is surjective.

Proposition 4. *Every set of at most $K = r^m - 1$ distinct points of X is linearly independent.*

Proof. Choose lifts $u_1, \ldots, u_s$ on C_D of $s \le K$ distinct image points. The lifts are distinct. A linear dependence among their images would give

$$\sum_{i=1}^{s} \alpha_i u_i = \lambda z.$$

If $\lambda \neq 0$, this expresses z using at most $s \le r^m - 1$ curve points, contrary to (3). If $\lambda = 0$, independence of at most $D+1$ distinct rational-normal-curve points forces every $\alpha_i = 0$. Thus no nontrivial dependence exists. $\square$

Now let $p = [\pi(e_1)]$. Since $\pi(z) = 0$,

$$\pi(e_1) = \pi(b) = \sum_{i=1}^{r} \pi(c_D(a_i)). \tag{4}$$

The r image points in this sum are distinct and independent. Distinctness also follows directly from (3): identifying two curve points would put z in their span and give rank at most two. In particular, the sum in (4) is nonzero.

This expression has minimal length r. Otherwise a shorter expression and (4) would give two different expressions in the span of at most $2r-1$ distinct points of X. Because $2r-1 \le r^m - 1 = K$, Proposition 4 makes that union independent. Uniqueness of basis coordinates would force all r original points to appear in the shorter expression, a contradiction. Thus $\mathrm{R}_X(p) = r$.

For border rank, as $t \to 0$ we have

$$\frac{\pi(c_D(t)) - \pi(c_D(0))}{t} = \pi(e_1) + \sum_{j=2}^{D} t^{j-1}\pi(e_j) \longrightarrow \pi(e_1).$$

For all sufficiently small nonzero t this is a nonzero vector of X-rank at most two. Its projective limit is p, so $\underline{\mathrm{R}}_X(p) \le 2$. Since $p \notin X$ and X is closed, the border rank is not one. Therefore $\underline{\mathrm{R}}_X(p) = 2$.

Finally, for $1 \le k \le m$ we have $r^k - 1 \le K$. Applying Lemma 3 proves all rank equalities in Theorem 1.

3.1 The curve is smooth

For completeness, the construction is not relying on singularities of X. In fact,

$$\underline{\mathrm{R}}_{C_D}([z]) = r + 2. \tag{5}$$

The upper bound is the sum of a tangent vector, of border rank two, and r curve vectors. To prove the lower bound, note that $D \ge 2r+2$ and form the $(r+2) \times (r+2)$ Hankel matrix

$$H = (z_{i+j})_{0 \le i,j \le r+1}, \qquad z_j = \delta_{j1} - \sum_{\nu=1}^{r} a_\nu^j.$$

Every curve vector maps to a matrix of rank at most one under this linear map, including the point at infinity. Hence a nonzero $(r+2)$-minor is a border-rank lower bound.

Put $u(a) = (1, a, \dots, a^{r+1})^\mathsf{T}$ and

$$B = \begin{bmatrix} u(0) & u'(0) & u(a_1) & \cdots & u(a_r) \end{bmatrix}, \qquad M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \oplus (-I_r).$$

Directly, $H = BMB^\mathsf{T}$. The square matrix B is invertible: a polynomial of degree at most $r+1$ in the kernel of B^T would have a double zero at zero and zeros at all a_i, and hence would be divisible by a polynomial of degree $r+2$. It must be zero. Thus H is invertible, proving (5).

In particular, the projection center lies on neither a secant nor a tangent line of C_D. The projection is injective on C_D and its differential is injective at every point: a failure of either assertion would place the center on the corresponding secant or tangent line. A proper injective immersion of this smooth complex projective curve is an embedding. Thus X is a smooth rational curve, as claimed.

4 An explicit counterexample in twelve coordinates

Take $r = 3$, $m = 2$, $D = 12$, and $a_1 = 1, a_2 = 2, a_3 = 3$. For $0 \le j \le 12$, set $S_j = 1 + 2^j + 3^j$, and use the coordinate index set

$$J = \{0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}.$$

Here $z_1 = 1 - S_1 = -5$. A quotient map with kernel $\mathbb{C}z$ is

$$P : E_{12} \longrightarrow \mathbb{C}^{12}, \qquad (P(v))_j = 5v_j - S_j v_1 \quad (j \in J).$$

It has rank twelve since its columns with indices in J form $5I_{12}$, and $P(z) = 0$. The curve is the image of the following explicit homogeneous parametrization:

$$[s : t] \longmapsto \left[\left(5s^{12-j}t^j - S_j s^{11} t \right)_{j \in J} \right]. \tag{6}$$

The common-factor-free, base-point-free property also follows from the quotient construction. Directly, at $s = 0$ the last coordinate is $5t^{12} \neq 0$; on the chart $s = 1$, vanishing of the first coordinate forces $t = 5/3$, at which the $j = 2$ coordinate is $-85/9 \neq 0$.

On this affine chart write $g(t) = (5t^j - S_j t)_{j \in J}$. A representative of the required point is

$$\begin{aligned} v = P(e_1) &= (-S_j)_{j \in J} \\ &= (-3, -14, -36, -98, -276, -794, -2316, -6818, \\ &\qquad\qquad -20196, -60074, -179196, -535538). \end{aligned}$$

The identities

$$g'(0) = v, \qquad g(1) + g(2) + g(3) = v$$

exhibit, respectively, its border-two degeneration and its three-term expression. The preceding proofs give

$$\boxed{\underline{\mathrm{R}}_X([v]) = 2, \qquad \mathrm{R}_X([v]) = 3, \qquad \mathrm{R}_{X \times X}([v \otimes v]) = 9.}$$

An explicit nine-term decomposition is

$$v \otimes v = \sum_{a=1}^{3} \sum_{b=1}^{3} g(a) \otimes g(b).$$

The lower bound applies to arbitrary complex decompositions on the whole curve, not only to decompositions using these three parameter values.

The 5×5 center Hankel determinant, using coordinates $z_0, \ldots, z_8$, equals 5184. This provides an additional exact check that the center has border rank at least five and that the example is a smooth projection.

5 Why there is no conflict with eventual strict submultiplicativity

The known eventual-power statement in [2, Section 1] concerns sufficiently large powers, not the square. There is an elementary eventual-saving bound for the present examples. Put

$$q(t) = \pi(c_D(t) - c_D(0)) = tv + t^2\pi(e_2) + \cdots + t^D\pi(e_D).$$

For an integer $k \geq 1$, set $L = k(D-1)+1$. Extracting the coefficient of t^k by a roots-of-unity sum gives

$$v^{\otimes k} = \frac{1}{L}\sum_{\zeta^L=1} \zeta^{-k} q(\zeta)^{\otimes k}.$$

The degrees of $q(t)^{\otimes k}$ lie between k and kD, so no other exponent is congruent to k modulo L. Each $q(\zeta)$ is a difference of two curve vectors. Therefore

$$\mathrm{R}_{X^{\times k}}(p^{\otimes k}) \leq (k(D-1)+1)2^k.$$

For fixed D and $r > 2$, this is less than r^k for all sufficiently large k. For the explicit $D = 12$ example, it already proves a saving at $k = 13$, since $144 \cdot 2^{13} = 1{,}179{,}648 < 3^{13} = 1{,}594{,}323$. Thus the examples delay a saving; they do not prevent one at every power.

6 Verification and scope

The exact-arithmetic script `verify.py` checks the coordinates, quotient rank and kernel, Hankel determinant, nine-term expression, and interpolation identities. It does not replace the universal independence and rank lower-bound proofs.

This counterexample answers the universal implication in TR-27. Arbitrary finite delay strengthens the same result; it is not a second problem claimed solved. The Segre and Veronese cases are not settled here.

References

[1] OpenProblemsInNLA, "TR-27—Border-rank deficiency forcing strict submultiplicativity at the tensor square," problem statement, checked September 11, 2026. `https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/tensor-computations/TR-27`.

[2] E. Ballico, A. Bernardi, F. Gesmundo, A. Oneto, and E. Ventura, "Geometric conditions for strict submultiplicativity of rank and border rank," *Annali di Matematica Pura ed Applicata* **200** (2021), 187–210. Conjecture 1.1. `https://arxiv.org/abs/1909.03811`.

[3] A. Oneto and E. Ventura, "Ranks of tensors: geometry and applications," *Bollettino dell'Unione Matematica Italiana* **18** (2025), 751–778. Conjecture 4.9. `https://doi.org/10.1007/s40574-025-00472-9`.